Newton School of Technology, Rishihood University

B.Tech Computer Science and Artificial Intelligence

Project Report

Card-Fraud Triage: GMM Density Scoring vs Autoencoder Anomaly Detector

Phase 2: Deep Learning Evaluation

Autoencoder anomaly detection and hybrid ML + DL density models

Project 26 | Team ID: T068 | Section: B

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Abstract

In Phase 1 we showed that a Gaussian Mixture Model (GMM), fitted only to legitimate card transactions after a quantile transform, is a strong label-free fraud ranker (test PR-AUC 0.686). We also showed that it works only after the inputs are forced to look Gaussian, and that the Expectation–Maximisation (EM) fit is unstable in high dimensions. In Phase 2 we ask whether *learned non-linear* models can close this gap.

We train undercomplete and denoising autoencoders only on legitimate transactions and score each transaction by its reconstruction error. We run 27 ablation variants (two seeds each) covering input scaling, bottleneck size, depth, corruption type and strength, dropout, weight decay, activation function, amount of training data and label contamination. The factors interact strongly, so the final networks are chosen among validated candidates rather than by stacking every individually helpful change. The denoising autoencoder reaches a test PR-AUC of 0.661. This is statistically on par with the GMM (paired bootstrap difference -0.025 [-0.055, +0.001]), and the model is stable across seeds. The plain undercomplete autoencoder is significantly worse (0.618).

We then test five ML + DL hybrids, all fitted in two stages without leakage: latent-space GMMs in the style of DAGMM, a reconstruction-augmented GMM, late score fusion, and semi-supervised stacking. None significantly beats the GMM. Our measurements show why: the autoencoder and the GMM find the same frauds, and the encoder's latent code discards the anomaly evidence, which lives in the reconstruction residual. A label-efficiency study shows that the label-free GMM is worth about 17 labelled frauds, and that a supervised model trained on all labels is significantly better. We therefore recommend deploying the GMM first and moving to a supervised model once a few dozen frauds have been confirmed.

Keywords: fraud detection, anomaly detection, autoencoder, denoising autoencoder, Gaussian mixture model, hybrid models, ablation study, label efficiency.

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1 Introduction

1.1 Recap of Phase 1

Phase 1 framed card-fraud triage as novelty detection. A GMM fitted by EM to *legitimate* transactions ranks every transaction by its negative log-likelihood $-\log p(\mathbf{x})$, so that analysts review the least "normal-looking" transactions first. With a quantile (marginal Gaussianisation) transform and $K = 4$ full-covariance components, the GMM reached a test PR-AUC of 0.686. It caught 75.7% of test frauds at a 0.1% false-positive budget, and 90% of its top-50 alerts were fraud.

1.2 The gap left by Phase 1

Phase 1 also measured three weaknesses of the GMM:

1. **Shape assumption.** The GMM works only after every feature is forced to look Gaussian. Each component is a fixed ellipsoid, and adding components did not help.
2. **Instability in high dimensions.** Each full-covariance component has 434 parameters in 28 dimensions, the fitted covariances are close to singular, and EM's solution depends on its random start.
3. **Model selection.** AIC and BIC keep asking for more components without any gain in detection quality.

1.3 Objectives of Phase 2

1. Train autoencoders (undercomplete and denoising) on legitimate transactions only, as named in the project brief, and score transactions by reconstruction error.
2. Justify every architectural and regularisation choice with ablation studies.
3. Compare the autoencoders with the Phase-1 GMM fairly: same rows, same split, same metrics, thresholds from validation, and paired statistical tests.
4. Design and evaluate hybrid ML + DL approaches, and explain when and why they add value.
5. Give a practical triage recommendation.

1.4 Contributions

- A fully ablated autoencoder study (27 variants, two seeds each), with a candidate-based selection procedure that handles interacting factors.
- A diagnosis of the relationship between reconstruction error and Gaussian density, tested empirically.
- A controlled hybrid study with leakage-free two-stage fitting: DAGMM-style latent den-

sities, a reconstruction-augmented GMM, late fusion, and a semi-supervised stacking model with a label-efficiency analysis.

- Multi-seed variance and paired bootstrap significance tests for every headline comparison.

2 Deep Learning Literature Review

2.1 Autoencoders and their relation to PCA

An autoencoder is a neural network trained to reproduce its input through a narrow intermediate layer. Baldi and Hornik [1] proved that a *linear* autoencoder trained with squared error has no spurious local minima and that its optimum spans the principal subspace of the data. A linear autoencoder is therefore equivalent to PCA. Hinton and Salakhutdinov [2] showed that deep non-linear autoencoders learn much better low-dimensional codes than PCA. Goodfellow, Bengio and Courville [3] give a textbook treatment of undercomplete, regularised and denoising autoencoders.

2.2 Reconstruction error as an anomaly score

Sakurada and Yairi [4] train autoencoders on normal data and use reconstruction error as the anomaly score. They show that this captures non-linear structure that linear PCA misses. An and Cho [5] propose a variational autoencoder with a "reconstruction probability" score. The surveys of Chalapathy and Chawla [6] and Pang *et al.* [7] identify reconstruction-based models as the dominant family of deep anomaly detectors. They also point out their main weakness: a network that generalises well can reconstruct anomalies too. Pumsirirat and Yan [8] apply autoencoders to credit-card fraud.

2.3 Denoising and regularisation

Vincent *et al.* [9, 10] introduced denoising autoencoders, which reconstruct a clean input from a corrupted copy. This forces robust features and prevents the network from learning the identity map. Alain and Bengio [11] showed that, for small Gaussian noise, the optimal denoising reconstruction satisfies $r(\mathbf{x}) - \mathbf{x} \propto \sigma^2 \nabla_{\mathbf{x}} \log p(\mathbf{x})$. A denoising autoencoder therefore implicitly learns the gradient of the log-density of the data, which links the reconstruction view of Phase 2 to the density view of Phase 1. Our training also uses dropout [12], the Adam optimiser [13] and ELU activations [14], each of which we ablate.

2.4 Hybrid deep and density models

DAGMM [15] couples an autoencoder with a GMM fitted on the concatenation of the latent code and two reconstruction features (relative Euclidean distance and cosine similarity). It trains both jointly and outperforms stand-alone autoencoders and GMMs on several benchmarks. Deep SVDD [16] learns a representation in which normal data falls inside a minimum-volume hypersphere. For card fraud, Carcillo *et al.* [17] combine unsupervised outlier scores with supervised

classifiers.

2.5 Summary

Table 1: Deep-learning literature and how it shapes our Phase-2 design.

Work	Contribution	How we use it
Baldi & Hornik [1]	Linear autoencoder = PCA	Ablation of the activation function (linear case)
Sakurada & Yairi [4]	Reconstruction error as anomaly score	Phase-2 baseline detector
Vincent *et al.* [9]	Denoising autoencoder	Denoising variant; noise ablation
Alain & Bengio [11]	Denoising learns the density gradient	Link between the GMM and the autoencoder
Zong *et al.* [15] (DAGMM)	GMM on latent code + reconstruction features	Hybrids H1 and H2
Carcillo *et al.* [17]	Unsupervised scores as features for supervised models	Semi-supervised hybrid H5
Pang *et al.* [7]	Reconstruction models can over-generalise	Diagnosis of over-generalisation

3 Data, Protocol and DL-specific Data Handling

3.1 Shared protocol

All Phase-2 models use exactly the same data and protocol as Phase 1:

- 1,081 duplicate rows are removed before splitting;
- rows are ordered by time and split into train (60%), validation (20%) and test (20%) blocks;
- models are fitted on legitimate training rows only;
- hyper-parameters and thresholds are chosen on validation;
- the test block (74 frauds, base rate 0.130%) is used once.

The input features are the Phase-1 feature set (V1–V28, $d = 28$).

3.2 Data handling for the deep models

- **Legitimate-only training.** The fraud label is used once, to remove the 342 training frauds. We also measure what happens when they are left in (ablation A7).
- **Label-free early stopping.** A random 10% of the legitimate training rows is held out and used only to monitor the reconstruction loss. No fraud label is involved in training or early stopping.

- **Input scaling.** Each model gets its own validated preprocessing. The GMM uses the quantile transform from Phase 1. For the autoencoder the scaler is chosen by ablation (A0); every scaler is fitted on legitimate training rows only.
- **Leakage control for hybrids.** Second-stage models are never fitted on rows that the autoencoder was trained on (Section 4.6).

4 Methodology

4.1 Autoencoder

An encoder $f_\theta : \mathbb{R}^d \to \mathbb{R}^m$ and a decoder $g_\phi : \mathbb{R}^m \to \mathbb{R}^d$ with $m < d$ are trained on legitimate rows to minimise the mean squared reconstruction error plus an L2 penalty:

$$\mathscr{L}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{x}_i - g_\phi(f_\theta(\tilde{\mathbf{x}}_i))\|_2^2 + \lambda \|\mathbf{W}\|_2^2. \tag{1}$$

For the undercomplete autoencoder $\tilde{\mathbf{x}}_i = \mathbf{x}_i$. The bottleneck ($m < d$) stops the network from learning the identity map: it must keep only the structure shared by legitimate transactions, the (possibly curved) manifold on which they lie.

4.2 Denoising autoencoder

The denoising autoencoder (DAE) trains on corrupted inputs and reconstructs the *clean* input. We consider two corruptions:

$$\text{Gaussian noise:} \quad \tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \tag{2}$$

$$\text{Masking:} \quad \tilde{x}_j = x_j \, b_j, \qquad b_j \sim \text{Bernoulli}(1 - p). \tag{3}$$

The network must learn to pull points back onto the data manifold. This acts as a strong regulariser, and by the result of Alain and Bengio [11] it ties the reconstruction to the gradient of the data's log-density. At test time no noise is added.

4.3 Anomaly score

Every transaction is scored by its reconstruction error on the clean input,

$$e(\mathbf{x}) = \frac{1}{d} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2, \qquad \hat{\mathbf{x}} = g_\phi(f_\theta(\mathbf{x})). \tag{4}$$

Thresholds are chosen on validation exactly as in Phase 1.

4.4 Architecture

The default network is a symmetric multilayer perceptron (Fig. 1). Table 2 lists the design choices and their reasons.

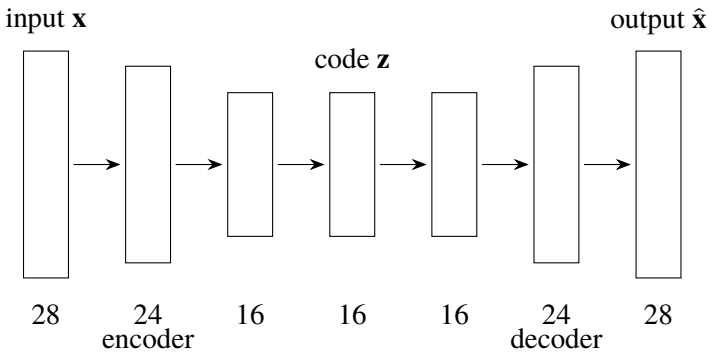


Figure 1: Selected autoencoder architecture ($d = 28$ inputs, hidden layers of 24 and 16 units, a 16-unit code). Numbers give the width of each layer. The denoising version uses the same network with Gaussian input noise during training.

Table 2: Architecture and training choices.

Choice	**Value**	**Reason**
Encoder	$d \to 24 \to 16 \to m$	gradual compression; m chosen by ablation (A1)
Decoder	mirror of the encoder, linear output	standardised inputs are unbounded, so no sigmoid or tanh at the output
Code layer	linear	lets the code use the whole real line
Activation	ELU	smooth and non-zero for negative inputs (no "dead" units); compared in A6
Loss	mean squared error	matches the reconstruction score
Regularisation	bottleneck, input noise, L2 weight decay, dropout, early stopping	each ablated (A1, A3–A5)
Optimiser	Adam, learning rate 2×10^{-3}, batch 2048	standard and stable for small networks
Schedule	ReduceLROnPlateau (factor 0.5, patience 3), at most 200 epochs	lowers the learning rate when the holdout loss stalls
Early stopping	patience 10 on a label-free 10% holdout	best weights are restored

Algorithm 1 Training an autoencoder on legitimate transactions

Require: legitimate training rows X; configuration (architecture, noise σ or masking p, weight decay λ)

Split X at random into a training part (90%) and a label-free holdout H (10%)
for epoch $= 1, \ldots, 200$ **do**
  for each mini-batch B of the training part **do**
    $\tilde{B} \leftarrow \text{corrupt}(B)$ ▷ identity for the undercomplete autoencoder
    $\hat{B} \leftarrow g_\phi(f_\theta(\tilde{B}))$; loss $\leftarrow \|\hat{B} - B\|^2 / |B|$
    update θ, ϕ with Adam (L2 weight decay λ)
  end for
  holdout loss $\leftarrow$ mean squared error of the clean reconstruction of H
  keep the weights with the lowest holdout loss; stop after 10 epochs without improvement
end for
return best weights; score $e(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 / d$

4.5 Why reconstruction error may differ from a Gaussian density

Let the covariance of legitimate data have eigenpairs $(\lambda_j, \mathbf{u}_j)$. The Gaussian score (Mahalanobis distance) is $\sum_j \left(\mathbf{u}_j^\top (\mathbf{x} - \boldsymbol{\mu})\right)^2 / \lambda_j$. It strongly amplifies deviations along directions in which legitimate data barely varies (small λ_j). A linear autoencoder with code size m has error $\sum_{j>m} \left(\mathbf{u}_j^\top (\mathbf{x} - \boldsymbol{\mu})\right)^2$: the residual directions are weighted *equally*, regardless of how tightly legitimate data is constrained along them. A non-linear autoencoder inherits this equal weighting in its residual. It may also partly reconstruct anomalies (over-generalisation [7]). We test both hypotheses in Section 5.7.

4.6 Hybrid designs

We evaluate five ways of combining the GMM (machine learning) with the autoencoder (deep learning). Each tests one idea (Table 3, Fig. 2).

Table 3: Hybrid designs.

ID	Stage-2 model	Idea
H0	GMM on $\mathbf{x}$ (same rows)	reference: the plain GMM fitted on the same rows as the hybrids
H1	GMM on $\mathbf{z}$	density estimation may be easier in a low-dimensional learned space (DAGMM idea [15])
H2	GMM on $[\mathbf{z}, \log e, \cos(\mathbf{x}, \hat{\mathbf{x}})]$	combine *where* a point is on the manifold with *how far* it is from it (DAGMM)
H3	GMM on $[\mathbf{x}, \log e]$	keep all input information and add the learned error as an extra dimension
H4	rank average of GMM and AE scores	naive late fusion, the baseline any real hybrid must beat
H5	gradient boosting on $[\mathbf{x}, \text{GMM score}, \log e]$	semi-supervised: use the few frauds analysts have labelled [17]

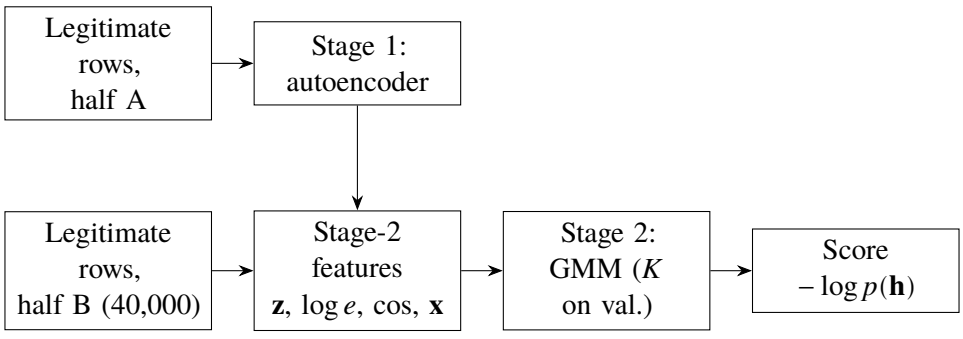


Figure 2: Two-stage fitting of the unsupervised hybrids (H1–H3). The autoencoder is trained on half A; every second-stage GMM is fitted on half B, rows the autoencoder never saw.

Leakage control. On the rows it was trained on, an autoencoder's reconstruction error is unrealistically small. A second-stage model fitted on those rows would learn a distorted "normal" error

level. For H1–H4 the legitimate training rows are therefore split at random into two halves. Half A trains the autoencoder, and a random 40,000 rows of half B fit every second-stage GMM, together with the H0 reference on the same rows. The whole procedure is repeated for three seeds, and for each second-stage GMM, $K \in \{1, 2, 3, 4\}$ is chosen on validation. For H5, GMM and autoencoder scores of training rows are computed *out of fold*: in a 2-fold cross-fitting, each half is scored by models fitted on the other half. The supervised model is trained with a varying number of labelled frauds (5 random draws per budget).

Why two-stage instead of joint training? DAGMM trains both parts jointly, which is known to be fragile: the GMM energy term can pull the encoder into degenerate solutions. Two-stage fitting is stable, and each stage can be validated separately. Joint training is left for future work.

4.7 Evaluation and statistical validation

We use the Phase-1 metrics:

- PR-AUC, with a 95% stratified bootstrap interval;
- recall at false-positive budgets of 0.1%, 0.5% and 1% (thresholds from validation);
- precision at review capacity (P@k).

To decide whether two models really differ, we use a *paired* stratified bootstrap. We resample the test set 1,000 times, compute PR-AUC(A) − PR-AUC(B) on each identical resample, and report the 95% interval of the difference together with the share of resamples in which A is not better than B. Every development decision uses validation PR-AUC averaged over seeds.

4.8 Experimental setup

Table 4: Experimental setup.

Item	Setting
Ablations	27 variants × 2 seeds; one factor changed at a time from the default network
Final models	3 seeds each; test scores averaged over seeds
Hybrids	3 seeds; second-stage $K \in \{1, 2, 3, 4\}$ chosen on validation
Label efficiency	6, 17, 34, 86 and 342 labelled frauds; 5 random draws per budget
Gradient boosting	300 iterations, learning rate 0.05, 15 leaves, L2 regularisation 1.0
Bootstrap	1,000 stratified resamples (paired for model comparisons)
Software	Python 3, PyTorch [18], scikit-learn [19]
Hardware	Google Colab with an NVIDIA T4 GPU; about 35 s per autoencoder training run

5 Experiments and Results

5.1 Ablation studies

Fig. 3 shows all ablation groups, and Table 14 (Appendix A) lists every variant. The default network scores only 0.078 ± 0.021 on validation. The main findings are:

1. **Bottleneck size is the largest single effect (A1).** Codes of 2–8 units are poor, and $m = 16$ is by far the best variant (≈ 0.44). The V-features are principal components with fairly evenly spread variance, so legitimate behaviour has no very low-dimensional structure. A tight code cannot rebuild *legitimate* rows well, and their large errors drown the fraud signal.

2. **Non-linearity is essential (A6).** The linear autoencoder, which is equivalent to PCA [1], is the worst variant (≈ 0.03). A network with no hidden layer also fails (A2).

3. **Denoising helps when the noise is right (A3, A4).** Gaussian noise of $\sigma = 0.2$ gives a large gain; smaller noise does nothing and $\sigma = 0.4$ is too strong. Masking noise is weaker than Gaussian noise here.

4. **Regularisation (A5).** Dropout *hurts*, because it makes reconstructions noisier, and that noise is exactly what we score. Stronger L2 weight decay (10^{-3}) helps by giving a smoother mapping that generalises less to unusual inputs.

5. **Inputs (A0).** Unlike the GMM, the autoencoder does not benefit from quantile inputs, which clip the extreme values that expose fraud. Robust scaling is slightly better than standard scaling on its own.

6. **Training data (A7).** More legitimate data helps. Leaving the 0.2% of training frauds in the training set is catastrophic, because the network learns to rebuild those extreme points too. The legitimate-only protocol is essential for this model.

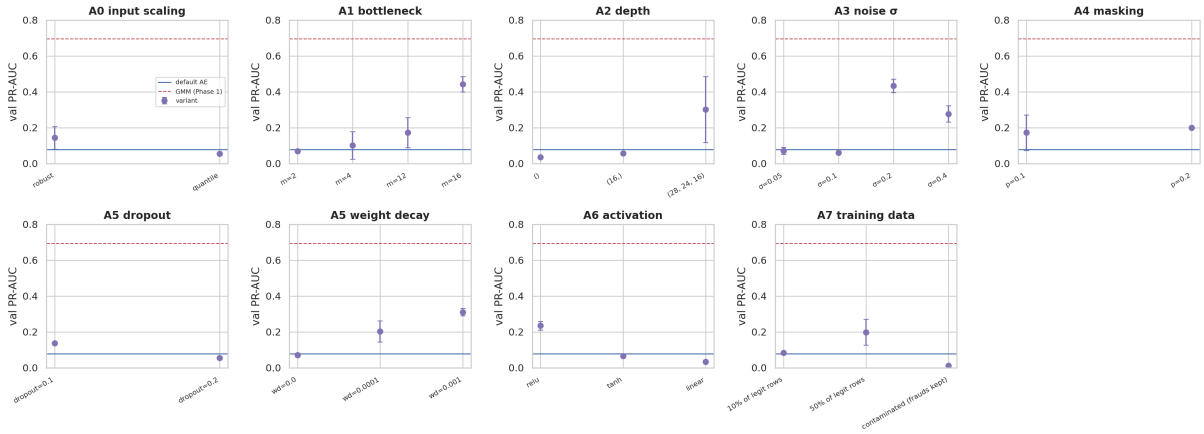


Figure 3: Autoencoder ablations: validation PR-AUC (mean $\pm$ std over 2 seeds) per factor. The solid line is the default network; the dashed line is the Phase-1 GMM.

5.2 Model selection with interacting factors

A natural next step would be to combine every change that helped on its own. We tried this: the combined network reached a validation PR-AUC of only about 0.02, worse than the default. The factors *interact*. For example, a wide code combined with dropout and a deeper stack behaves very differently from each change alone. We therefore compare a short list of candidates on validation (Table 5):

- the default network;
- each factor's best setting on its own;
- the two strongest changes combined;
- all improving changes combined.

The winner is the single best change, a 16-unit code with standard inputs. For the denoising autoencoder we add the best Gaussian-noise and the best masking setting to this network and again compare on validation. Gaussian noise $\sigma = 0.2$ wins clearly.

Table 5: Candidate configurations compared on validation (PR-AUC, mean $\pm$ std over 2 seeds).

Candidate	**Validation PR-AUC**	**Source**
default	0.078$\pm$0.021	ablation study
A1 bottleneck: m=16	0.442$\pm$0.042	ablation study
A5 weight decay: wd=0.001	0.311$\pm$0.021	ablation study
A2 depth: (28, 24, 16)	0.301$\pm$0.184	ablation study
A6 activation: relu	0.235$\pm$0.024	ablation study
A0 input scaling: robust	0.144$\pm$0.063	ablation study
A5 dropout: dropout=0.1	0.137$\pm$0.005	ablation study
top-2 combined	0.275$\pm$0.183	trained for selection
all improvements combined	0.024$\pm$0.000	trained for selection
chosen AE + σ=0.2	0.621$\pm$0.036	trained for selection
chosen AE + p=0.2	0.345$\pm$0.040	trained for selection

5.3 Training behaviour

Fig. 4 shows the training and holdout losses of the final networks. Both converge smoothly; early stopping on the label-free holdout restores the best epoch. For the denoising autoencoder the training loss is measured on corrupted inputs, so it stays above the clean holdout loss.

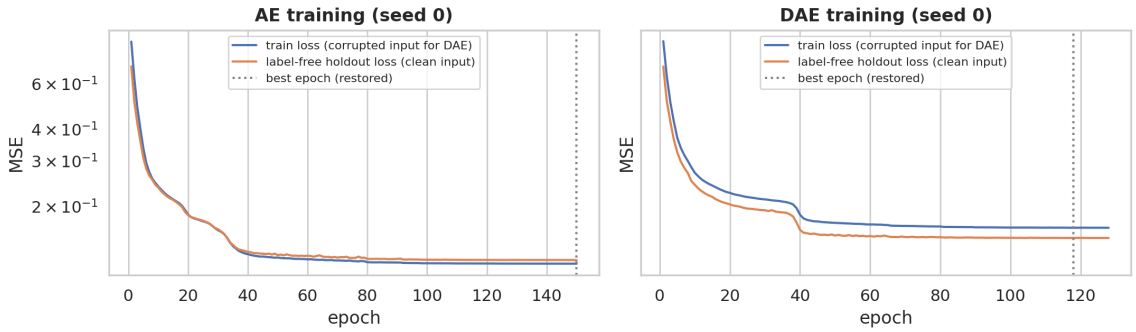


Figure 4: Training loss and label-free holdout loss of the final autoencoder (left) and denoising autoencoder (right), seed 0.

5.4 Autoencoders versus the Phase-1 GMM

Table 6 compares the final models on the test set (autoencoder scores averaged over three seeds), Table 7 gives the paired significance tests, and Fig. 5 shows the curves.

- The **denoising autoencoder** reaches PR-AUC 0.661. It is statistically on par with the GMM (0.686; difference -0.025 [-0.055, +0.001]) and stable across seeds (0.628 $\pm$ 0.013).
- The **undercomplete autoencoder** is significantly worse than the GMM (-0.068 [-0.107, -0.032]) and sensitive to the seed (0.569 $\pm$ 0.068).
- **Denoising improves on plain reconstruction** by +0.043 [+0.018, +0.075], acting as a strong regulariser [9].
- At strict budgets the GMM remains ahead. It catches 75.7% of frauds at a 0.1% false-positive rate against 70.3% for the denoising autoencoder, and its top-50 queue holds 90% fraud against 84%.

Table 6: Test results: Phase-1 GMM versus autoencoders (thresholds from validation; base rate 0.130%).

Model	PR-AUC	95% CI	ROC-AUC	Rec@0.1%	Rec@0.5%	Rec@1%	P@50	P@100
GMM (Phase 1)	0.686	[0.57, 0.81]	0.934	0.757	0.784	0.784	0.90	0.56
Undercomplete AE	0.618	[0.50, 0.74]	0.938	0.703	0.770	0.770	0.72	0.52
Denoising AE	0.661	[0.55, 0.78]	0.940	0.703	0.784	0.797	0.84	0.55

Table 7: Paired bootstrap tests on the test set (1,000 resamples).

Comparison	Δ PR-AUC	95% CI	P(A not better)
Undercomplete AE – GMM	-0.068	[-0.107, -0.032]	1.000
Denoising AE – GMM	-0.025	[-0.055, +0.001]	0.964
Denoising AE – Undercomplete AE	+0.043	[+0.018, +0.075]	0.000

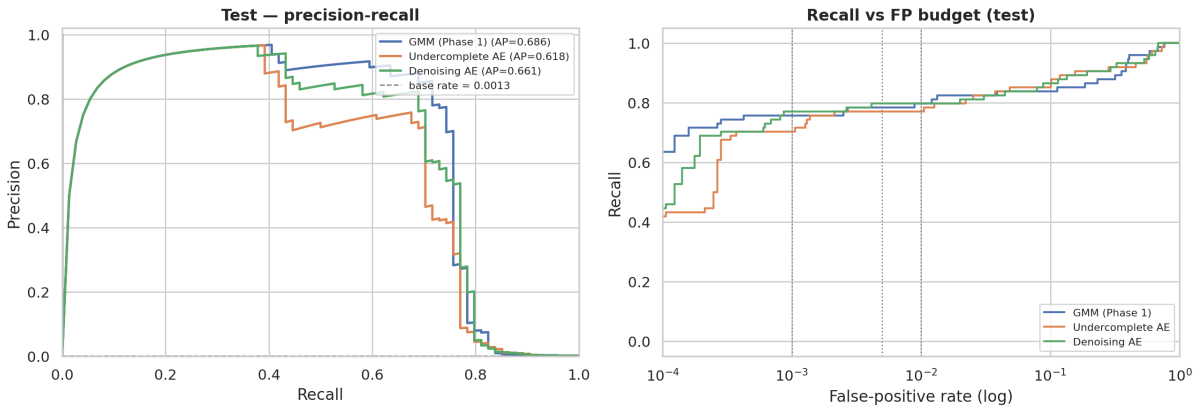


Figure 5: Test precision–recall curves (left) and recall versus false-positive rate (right) for the GMM and the two autoencoders.

5.5 Threshold analysis

Fig. 6 shows precision and recall against review workload for the three models. The curves almost coincide for large queues. The GMM is slightly better for small, high-precision queues, which matter most when analysts have little capacity.

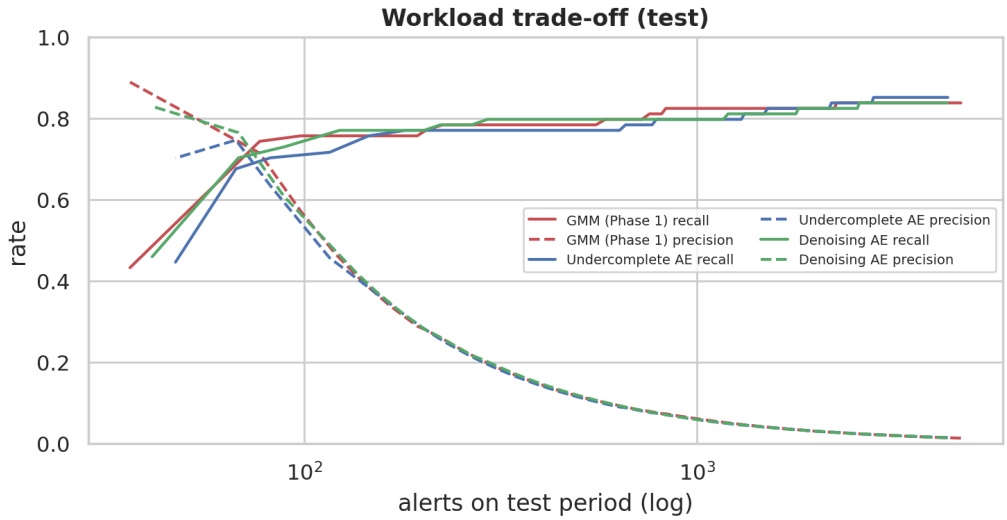


Figure 6: Precision (dashed) and recall (solid) versus number of alerts on the test period.

5.6 Interpretability: which features does the autoencoder fail to rebuild?

Reconstruction error decomposes over features, $e(\mathbf{x}) = \frac{1}{d}\sum_j (x_j - \hat{x}_j)^2$. Averaging each term over frauds and over legitimate rows shows which features the network cannot rebuild for fraud (Fig. 7). V17, V14 and V12 dominate, the same components that separate the classes in the exploratory analysis of Phase 1. An analyst can therefore be given a reason for each alert. Fig. 8 shows the score distribution and a 2-D projection of the latent codes.

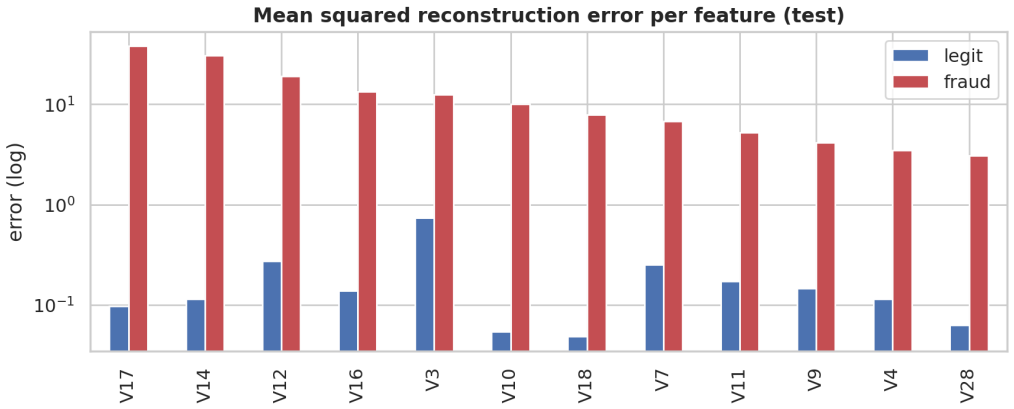


Figure 7: Mean squared reconstruction error per feature for legitimate and fraudulent test transactions (log scale).

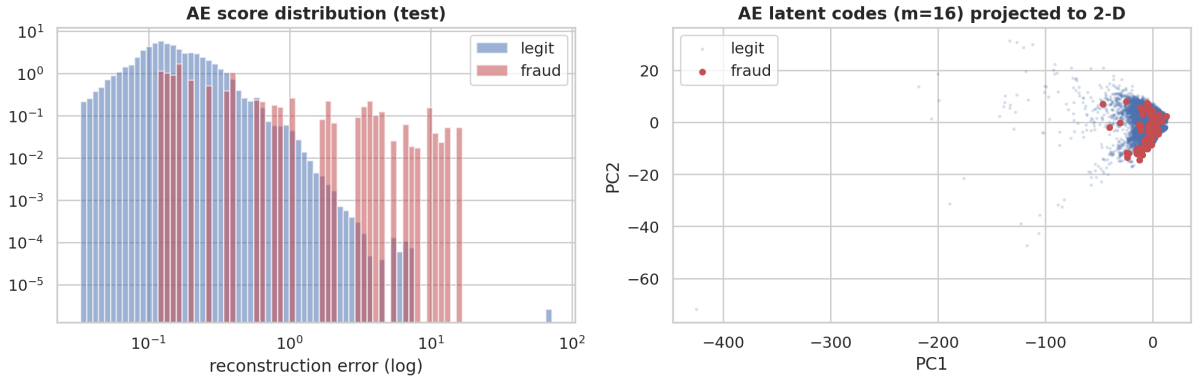


Figure 8: Autoencoder score distribution on test (left) and the latent codes projected to two dimensions (right).

5.7 Diagnosis

Three measurements test the hypotheses of Section 4.

1. **Agreement.** On frauds, the autoencoder error and the GMM score rank transactions almost identically (Spearman $\rho = 0.941$). On legitimate rows they agree much less ($\rho = 0.429$). The two models find the same frauds but raise *different* false alarms. A hybrid could therefore only hope to reduce false alarms, not to find new frauds.

2. **Direction weighting.** Re-scoring the same autoencoder's residuals with a Gaussian fitted to legitimate residuals (a Mahalanobis residual score) *lowers* test PR-AUC from 0.583 to 0.171. The residuals are themselves heavy-tailed, so the theoretically motivated fix fails for the same reason a GMM fails on raw heavy-tailed inputs. The hypothesis is not supported in this simple form.

3. **Over-generalisation.** The autoencoder reproduces 73.6% of a typical legitimate row's deviation from the mean, but only 31.7% of a typical fraud's. It does partly rebuild fraud, which limits its recall, but the gap is large enough to rank most frauds high.

5.8 Hybrid study

Table 8 reports each unsupervised hybrid at its best second-stage K (mean $\pm$ std over three seeds), and Fig. 9 shows the comparison and the effect of K.

- **H3** (GMM on raw features plus the log reconstruction error) ties the plain GMM fitted on the same rows (H0). Adding the learned error gives the mixture nothing it does not already know.
- **H4** (late fusion) is worse than the GMM alone. The stage-1 network, trained on only half the legitimate rows, is weak and unstable on its own, and averaging it in dilutes the GMM's ranking.
- **H1 and H2**, the DAGMM-style latent-space hybrids, fail almost completely.

Table 8: Unsupervised hybrids at the best second-stage K on validation (mean $\pm$ std over 3 seeds).

Model	K	Validation PR-AUC	Test PR-AUC
Phase-1 GMM (all legit rows)	4	0.695±0.000	0.686±0.000
H0 GMM on x (same rows)	2	0.688±0.001	0.697±0.013
H3 augmented GMM [x, log e]	2	0.686±0.002	0.697±0.013
H4 late fusion (rank avg)	–	0.342±0.186	0.524±0.137
AE alone (reconstruction error)	–	0.163±0.111	0.325±0.114
H2 DAGMM-style [z, log e, cos]	3	0.025±0.002	0.044±0.006
H1 latent GMM [z]	4	0.008±0.002	0.012±0.004

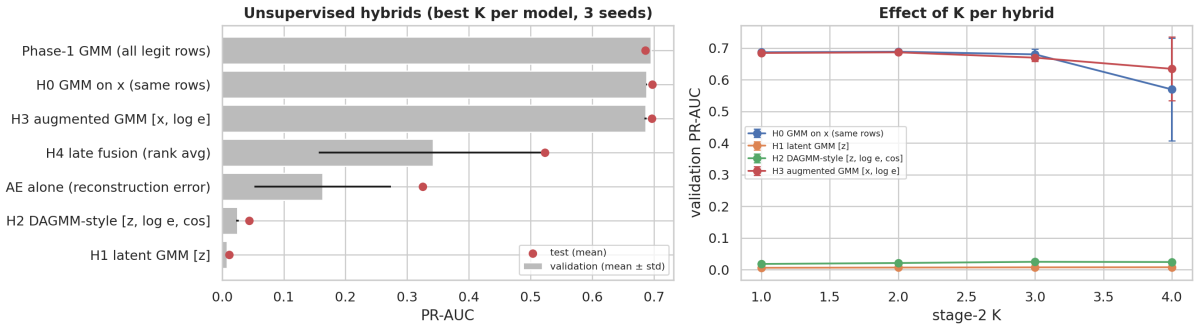


Figure 9: Hybrid ablation. Left: validation (bars, mean $\pm$ std) and test (dots) PR-AUC per design. Right: effect of the second-stage K.

5.9 Where does the anomaly information live?

Table 9 explains the failure of the latent-space hybrids. At a 1% false-positive rate the latent GMM flags only 4.1% of test frauds, while the GMM on the raw inputs flags 77.0% and the reconstruction error 77.0%. Fraud codes are barely larger than legitimate ones (mean $|\mathbf{z}|$ 3.52 against 2.91). An encoder trained only on legitimate data maps *any* input close to the region of normal codes. What exposes a fraud is that the decoder cannot rebuild it from that normal-

looking code. The evidence therefore lies in the reconstruction residual, not in the code, and a density model of the code cannot see it.

Table 9: Evidence on the test set (seed 0). "Above legit 99th pct" = share of frauds flagged at a 1% false-positive rate.

Measurement	Value
frauds above legit 99th pct — latent GMM on z (H1)	4.1%
frauds above legit 99th pct — GMM on x (H0)	77.0%
frauds above legit 99th pct — AE error	77.0%
mean |z| legit	2.908
mean |z| fraud	3.524
Spearman(AE error, GMM score) — test frauds	0.952
Spearman(AE error, GMM score) — test legit	0.384

5.10 Semi-supervised hybrid and label efficiency

Table 10 and Fig. 10 train gradient boosting on the training block with only n labelled frauds (five random draws per budget):

- with 6 labelled frauds the supervised model is worse than the label-free GMM and very unstable;
- with about 17 it matches the GMM;
- with 34 or more it is clearly better.

In other words, *the label-free GMM is worth roughly 17 labelled frauds*. Adding the GMM score and the reconstruction error as features (H5) does not change PR-AUC measurably at any budget. Gradient boosting already extracts the relevant structure from the raw features.

Table 10: Label efficiency: test PR-AUC (mean $\pm$ std over 5 random draws) against the number of labelled training frauds.

Labelled frauds	Raw features	Raw + GMM score + AE error (H5)
6	0.523$\pm$0.236	0.531$\pm$0.252
17	0.696$\pm$0.073	0.692$\pm$0.072
34	0.747$\pm$0.029	0.759$\pm$0.021
86	0.769$\pm$0.013	0.772$\pm$0.009
342	0.778$\pm$0.000	0.771$\pm$0.000

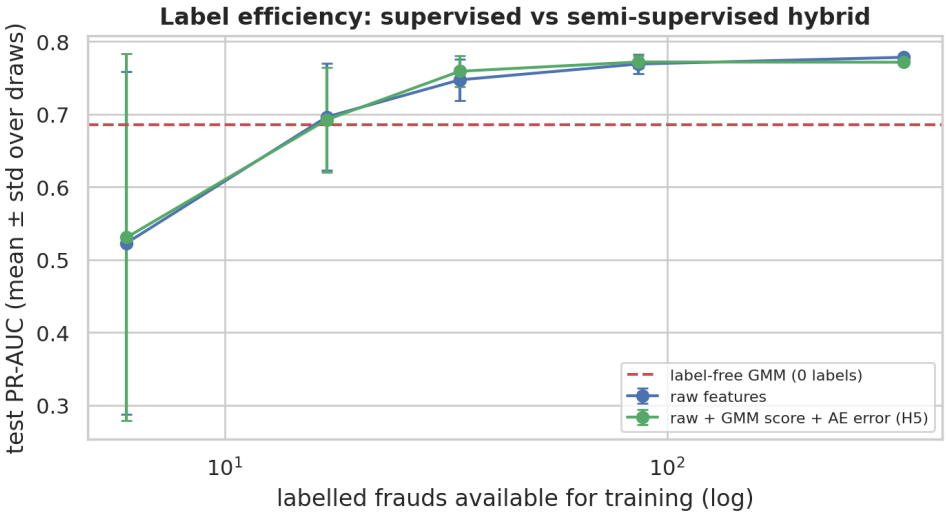


Figure 10: Label efficiency. The dashed line is the label-free GMM.

5.11 Final comparison

Table 11 and Fig. 11 compare all approaches on the test set, and Table 12 gives the paired tests. No unsupervised hybrid beats the Phase-1 GMM: the best one (H3) differs by +0.009 [-0.009, +0.023], which is not significant. Labels are what move the needle: supervised gradient boosting with all training labels beats the GMM by +0.092 [+0.026, +0.145]. The semi-supervised hybrid is no better than supervised alone (-0.007 [-0.045, +0.033]). Note that the "Stage-1 DAE alone" row is the autoencoder trained on half the rows for the hybrid study; the full-data denoising autoencoder of Table 6 scores 0.661.

Table 11: Final test comparison (thresholds from validation; base rate 0.130%). Models marked "uses labels" are references, not fair competitors to the label-free models.

Model	PR-AUC	95% CI	ROC-AUC	Rec@0.1%	Rec@0.5%	Rec@1%	P@50	P@100
GMM (Phase 1)	0.686	[0.57, 0.81]	0.934	0.757	0.784	0.784	0.90	0.56
Stage-1 DAE alone (half the rows)	0.301	[0.24, 0.40]	0.943	0.189	0.716	0.770	0.48	0.40
Hybrid H3 augmented GMM [x, log e]	0.695	[0.59, 0.81]	0.961	0.757	0.757	0.770	0.92	0.56
Supervised GBM, raw features (uses labels)	0.778	[0.69, 0.86]	0.976	0.784	0.838	0.865	0.96	0.58
Semi-supervised hybrid H5 (uses labels)	0.771	[0.68, 0.86]	0.978	0.757	0.811	0.865	0.94	0.56

Table 12: Paired bootstrap tests for the hybrid study (1,000 resamples).

Comparison	Δ PR-AUC	95% CI	P(A not better)
H3 augmented GMM [x, log e] − GMM	+0.009	[-0.009, +0.023]	0.325
Stage-1 DAE alone − GMM	-0.384	[-0.474, -0.299]	1.000
H5 − supervised GBM	-0.007	[-0.045, +0.033]	0.647
supervised GBM − GMM	+0.092	[+0.026, +0.145]	0.001

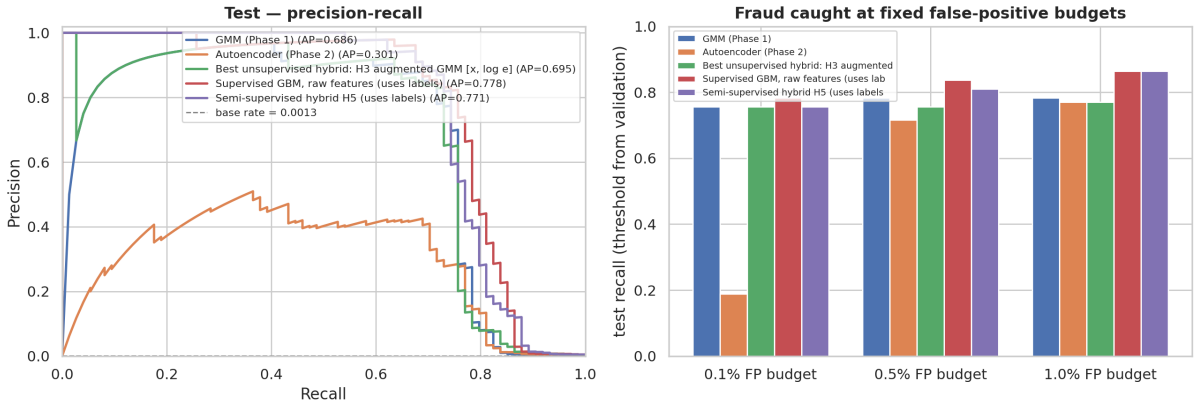


Figure 11: Final comparison on test: precision–recall curves (left) and recall at fixed false-positive budgets (right).

5.12 Triage recommendation

The recommended label-free system, chosen on validation, is the Phase-1 GMM. Table 13 lists its operating points on the 7.7-hour test period. A strict 0.1% budget produces about 12 alerts per hour, of which roughly 60% are fraud, and catches three quarters of all frauds. A larger queue adds very few frauds at a high cost in false alarms.

Table 13: Operating points of the recommended label-free model on the test period.

Policy	Alerts	Per hour	Caught	Precision	Legit flagged
FP budget 0.1%	94	12.3	56/74	0.596	38
FP budget 0.5%	221	28.9	58/74	0.262	163
FP budget 1.0%	399	52.1	58/74	0.145	341
review capacity top-50	50	6.5	45/74	0.900	5
review capacity top-100	100	13.1	56/74	0.560	44
review capacity top-200	200	26.1	57/74	0.285	143
review capacity top-500	500	65.3	58/74	0.116	442

6 Discussion

6.1 Technical validation

Every claim in this report rests on validation-based decisions and a single use of the test set. Headline comparisons are averaged over several seeds and checked with paired bootstrap tests. All second-stage models are fitted without leakage. The same pipeline run twice on Colab produced identical results.

6.2 Why the hybrids did not beat the GMM

On this dataset the quantile-transformed Gaussian density already captures what the autoencoder learns about fraud. Both models flag the same frauds, and they differ only on legitimate rows.

Hybrids that keep the raw inputs therefore tie the GMM, and hybrids that rely on the latent code lose the very information that identifies fraud. This is a useful negative result: it shows *where* the anomaly signal lives (in the residual, not the code), and it shows that DAGMM-style designs need care when the encoder is trained only on normal data.

6.3 Practical implications

For a bank starting a fraud-triage system, the results suggest a simple path:

1. deploy the label-free GMM first, with a threshold set by the review capacity;
2. collect analyst verdicts on the alerts;
3. once a few dozen frauds are confirmed, switch to (or add) a supervised model;
4. retrain both regularly, because fraud patterns drift.

6.4 Limitations

- The data covers two days, with 74 test frauds, so confidence intervals are wide (about ± 0.1 PR-AUC).
- The features are anonymised principal components, which limits feature engineering and interpretation.
- The hybrids are trained in two stages rather than jointly.
- The stage-1 autoencoder of the hybrid study is trained on half the legitimate rows, which weakens late fusion.

7 Conclusion and Future Work

Autoencoders trained only on legitimate transactions can detect card fraud, but their design must be validated carefully. The bottleneck size, the non-linearity and Gaussian input noise decide performance, and stacking individually helpful changes without re-validation fails. The selected denoising autoencoder matches the Phase-1 GMM within statistical noise and is the most stable deep model. No hybrid significantly improves on the GMM. The best one ties it, and DAGMM-style latent densities fail because the encoder maps fraud to normal-looking codes. Labels matter more than model combination: the label-free GMM is worth about 17 labelled frauds, and supervised boosting with all labels is significantly better.

Future work includes:

- end-to-end (joint) DAGMM training;
- flexible density models, such as normalising flows, on the reconstruction residual rather than the latent code;
- longer data with real concept drift;

- cost-sensitive thresholds that use transaction amounts.

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A Full Ablation Results

Table 14: All autoencoder ablation variants (validation PR-AUC, mean ± std over 2 seeds; mean number of epochs before early stopping).

Group	Setting	Validation PR-AUC	Epochs
default	default AE	0.078±0.021	200
A0 input scaling	robust	0.144±0.063	200
A0 input scaling	quantile	0.055±0.004	200
A1 bottleneck	m=2	0.068±0.006	150
A1 bottleneck	m=4	0.101±0.077	161
A1 bottleneck	m=12	0.172±0.083	182
A1 bottleneck	m=16	0.442±0.042	175
A2 depth	()	0.034±0.001	45
A2 depth	(16,)	0.056±0.002	168
A2 depth	(28, 24, 16)	0.301±0.184	187
A3 noise σ	σ=0.05	0.071±0.020	200
A3 noise σ	σ=0.1	0.060±0.008	180
A3 noise σ	σ=0.2	0.433±0.037	200
A3 noise σ	σ=0.4	0.276±0.045	198
A4 masking	p=0.1	0.172±0.099	152
A4 masking	p=0.2	0.199±0.002	106
A5 dropout	dropout=0.1	0.137±0.005	108
A5 dropout	dropout=0.2	0.055±0.003	75
A5 weight decay	wd=0.0	0.071±0.011	192
A5 weight decay	wd=0.0001	0.204±0.059	200
A5 weight decay	wd=0.001	0.311±0.021	200
A6 activation	relu	0.235±0.024	200
A6 activation	tanh	0.066±0.003	200
A6 activation	linear	0.034±0.001	18
A7 training data	10% of legit rows	0.083±0.004	200
A7 training data	50% of legit rows	0.198±0.072	200
A7 training data	contaminated (frauds kept)	0.012±0.003	200

B Reproducibility

- **Code.** Phase 2 uses notebooks 03 (autoencoders) and 04 (hybrid study). Both run top-to-bottom on Google Colab with a T4 GPU, download the dataset and verify its checksum.
- **Seeds.** Fixed seed lists for ablations (2 seeds), final models (3 seeds), hybrids (3 seeds) and label draws (5 draws).
- **Traceability.** All numbers in this report are generated from `results/*.json` into `numbers.tex`; all figures come from `figures/`.

- **Run time on Colab.** Notebook 03 takes about 35 minutes and notebook 04 about 15 minutes.

C Team Contributions

Table 15: Team T068 (Section B): members and primary responsibilities in Phase 2. Every member can explain the full pipeline.

Name	Enrollment No.	Primary responsibility (Phase 2)
Sharma Piyush	2401010437	DL data handling: legitimate-only training, label-free holdout, contamination study
Saransh Singh Tomar	2401010418	Input preprocessing for the autoencoder and the scaling ablation
Satwik Mani Tripathi	2401020056	Hybrid second stage: latent GMMs, augmented GMM and the evidence analysis
Gourav Tanwar	2401010173	Autoencoder and denoising autoencoder: architecture, ablations and model selection
Alok Kumar	2401010058	Evaluation, significance tests, label efficiency, report and presentation